

Solution of Tutorial-5
Engineering Mechanics: ME-101 (Semester-II, 2024)
Time: 7:55 – 8:50 AM

Q1. A person of mass $m_1 = 80$ kg kneeling on a plywood panel of mass $m_2 = 10$ kg wishes to push himself and the panel away from the right-hand wall (Figure 1). Determine a condition on μ_2 which would allow him to do this, as opposed to his slipping relative to a stationary panel. The two applicable coefficients of static friction are $\mu_1 = 0.60$ and μ_2 .

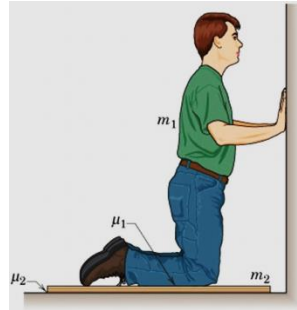
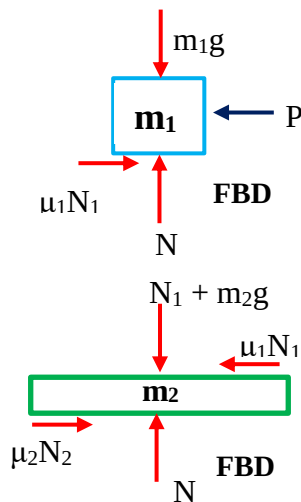


Figure 1

Solution of Q1:

Let us first draw the free body diagram for mass m_1 (of the person) and m_2 (of the plywood panel) of the assembly.



2 marks

From the free body diagram of mass m_1

3 marks

$$\sum F_y = 0 \text{ leads to } N_1 = m_1 g$$

1 mark

From the free body diagram of mass m_2

1 mark

$$\sum F_y = 0 \text{ leads to } N_2 = N_1 + m_2 g = (m_1 + m_2) g$$

$$\sum F_x = 0 \text{ leads to } \mu_1 N_1 = \mu_2 N_2 \Rightarrow \mu_2 = \mu_1 \times \frac{N_1}{N_2}$$

1 mark

$$\Rightarrow \mu_2 = 0.6 \times \frac{(m_1)}{(m_1 + m_2)} \Rightarrow \mu_2 = 0.6 \times \frac{80}{90}$$

1 mark

$\Rightarrow \mu_2 = 0.533$ (condition for the assembly to be at rest)

$\therefore \mu_2 < 0.533$ will ensure that the person will succeed in pushing himself and the plywood panel away from the right wall. 1 mark

Q2. The vertical position of 100 kg block is adjusted by the screw activated wedge (Figure 2). Calculate the moment M which must be applied to the handle of the screw to raise the block. The single threaded screw has square threads with mean diameter of 30 mm and advances 10 mm for each complete turn. The coefficient of friction for screw thread is 0.25 and all mating surfaces of the block and wedge is 0.4. Neglect friction at the ball joint A.

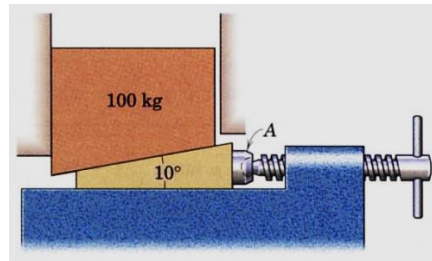


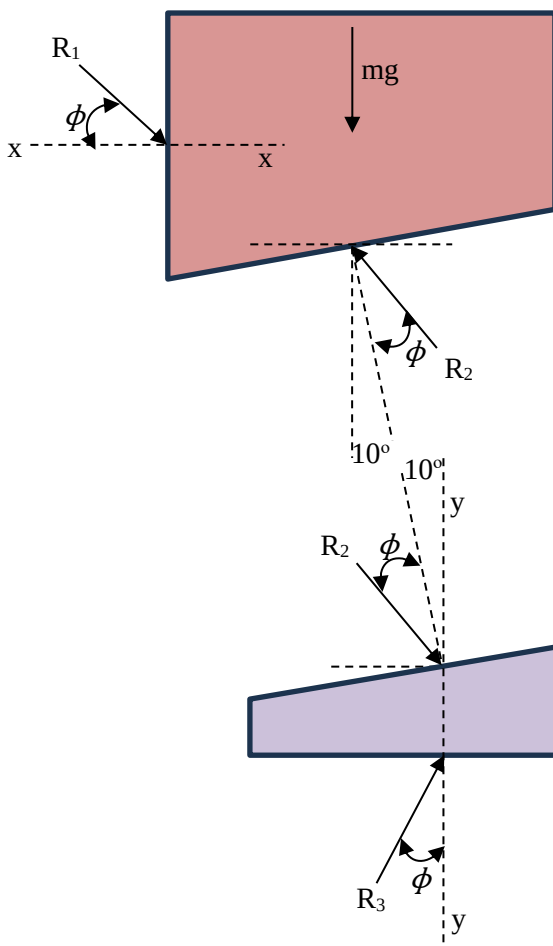
Figure 2

Solution of Q2:

Lead (advancement per revolution) (L) = 10 mm

Mean radius of the screw ($d/2$) = 15 mm

The free-body diagrams of the wedge and the block are drawn with the reactions R_1 , R_2 , and R_3 inclined with respect to their normal by the amount of the friction angles for impending motion.



2 marks

Free Body Diagrams for block and wedge

2 marks

The coefficient of friction for all mating surfaces of the block and wedge is given as 0.4.

$$\phi = \tan^{-1} \mu = \tan^{-1} 0.40 = 21.80^\circ$$

Consider free body diagram of the block and apply equilibrium of forces.

$$\sum F_x = 0: R_1 \cos \phi - R_2 \sin(\phi + 10^\circ) = 0$$

$$R_1 \cos 21.8^\circ - R_2 \sin(31.8^\circ) = 0 \text{ ----- (1)}$$

1 mark

$$\sum F_y = 0: R_1 \sin \phi + mg - R_2 \cos(\phi + 10^\circ) = 0$$

$$R_1 \sin 21.8^\circ + 981 - R_2 \cos(31.8^\circ) = 0 \text{ ----- (2)}$$

By solving equations (1) and (2)

$$R_1 = 875 \text{ N}$$

$$R_2 = 1535 \text{ N}$$

1 mark

Consider free body diagram of the wedge and apply equilibrium of forces.

$$\sum F_x = 0: R_2 \sin(\phi + 10^\circ) + R_3 \sin \phi - P = 0$$

$$R_2 \sin(31.8^\circ) + R_3 \sin 21.8^\circ - P = 0 \text{ ----- (3)}$$

$$\sum F_y = 0: R_2 \cos(\phi + 10^\circ) - R_3 \cos \phi = 0$$

$$1535 \times \cos(31.8^\circ) - R_3 \cos 21.8^\circ = 0 \text{ ----- (4)}$$

By solving equations (3) and (4)

$$R_3 = 1405 \text{ N}$$

$$P = 1331 \text{ N}$$

1 mark

$$\text{The helix angle } \alpha = \tan^{-1}(L/2\pi r)$$

$$\alpha = \tan^{-1}(10/2\pi 15) = 6.06^\circ$$

1 mark

$$\text{The angle of friction for the thread: } \phi' = \tan^{-1} 0.25 = 14.04^\circ$$

$$\phi' + \alpha = 20.09^\circ$$

1 mark

The moment M which must be applied to the handle of the screw to raise the block is given by.

$$M = Pr \tan(\phi' + \alpha) = 1331 \times (0.015) \tan 20.09^\circ = 7.30 \text{ Nm}$$

1 mark

Q3. The tape slides around the two fixed pegs of diameter 30 mm as shown in Figure 3 and is under the action of the horizontal tensions $T_1 = 40 \text{ N}$ and $T_2 = 160 \text{ N}$. Determine the coefficient of friction μ between the tape and the pegs.

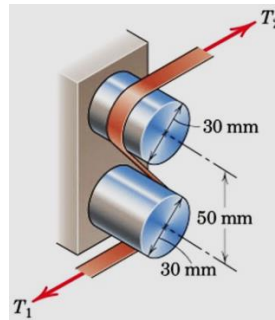
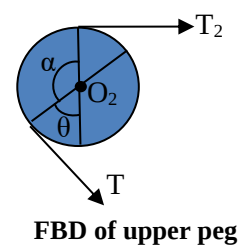
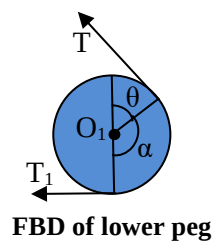
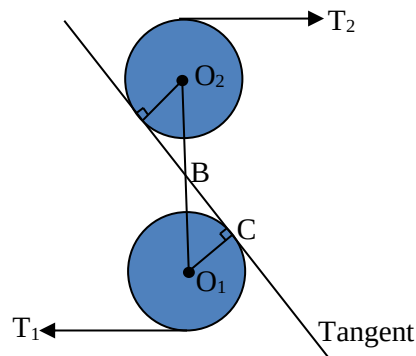


Figure 3

Solution of Q3:



3 marks

$$O_1B = 25 \text{ mm}$$

$$O_2C = 25 \text{ mm}$$

$$O_1C = 15 \text{ mm (radius of peg)}$$

$$\angle BO_1C = \theta = \cos^{-1}(15/25) = 53.13^\circ$$

1 mark

$$\text{Total angle of belt contact in radians: } \alpha = 180^\circ - \theta = 126.87^\circ = 2.214 \text{ radian}$$

$$\text{Friction coefficient} = \mu$$

$$\text{For lower peg: } \frac{T}{T_1} = e^{\mu\alpha}$$

2 marks

$$\text{For upper peg: } \frac{T_2}{T} = e^{\mu\alpha}$$

2 marks

$$\Rightarrow T_2 = T_1 e^{2\mu\alpha}$$

$$\Rightarrow 160 = 40 \times e^{2\mu \times 2.214}$$

$$\Rightarrow \mu = 0.313$$

2 marks

Q4. The system of two blocks, cable, and fixed pulley is initially at rest. The horizontal force P necessary to cause motion when (a) P is applied to the 10-kg block and (b) P is applied to the 20-kg block. [10 marks]

- a) Draw free body diagram for case a [1 mark]
- b) Calculate the normal reaction forces between the blocks and between the lower block and fixed surface. [2 marks]
- c) Determine force P on 10 kg block and tension T in the cable for case a [3marks]
- d) Draw free body diagram for case b [1 mark]
- e) Determine force P on 20 kg block and Tension T in the cable for case b [3 marks]

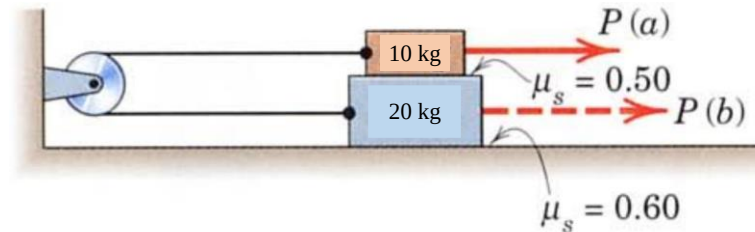


Figure 4

Solution of Q4:

Step 1: Free body diagram for case a

[1 mark]

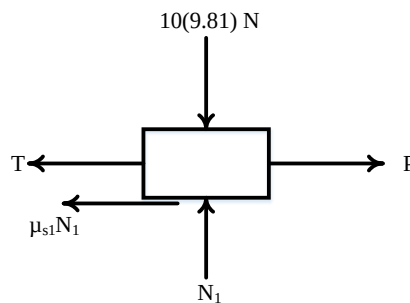


Figure 4.1

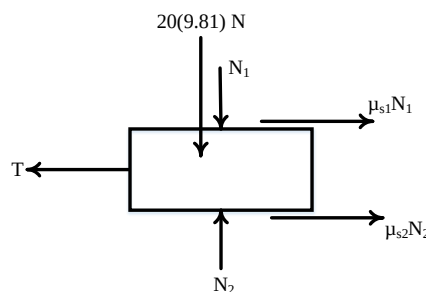


Figure 4.2

Coefficient of friction between 10 kg block and 20 kg block

$$\mu_{s_1} = 0.5$$

Coefficient of friction between 20 kg block and fixed surface

$$\mu_{s_2} = 0.6$$

Step 2: Calculation of normal reaction force**[2 marks]**

From figure 4.1, force balance equation in y- direction, taking upward direction force as positive and downward direction force as negative.

$$\begin{aligned}\sum F_y &= 0 \\ N_1 - 10(9.8) &= 0 \\ N_1 &= 98 \text{ N}\end{aligned}$$

From figure 4.2, force balance equation in y- direction, taking upward direction force as positive and downward direction force as negative.

$$\begin{aligned}\sum F_y &= 0 \\ N_2 - 20(9.8) - N_1 &= 0 \\ N_2 - 20(9.8) - 98 &= 0 \\ N_2 &= 294 \text{ N}\end{aligned}$$

Case a: When force P is applied on 10 kg block**Step 3: Applied force P on 10 kg block and Tension T in the cable for case a** **[3 marks]**

Force balance equation in x-direction, taking right direction force is positive and left direction force is negative.

From figure 4.1

$$\begin{aligned}\sum F_x &= 0 \\ P - T - \mu_{s1} N_1 &= 0 \\ P - T - 0.5(98) &= 0 \dots \dots \dots (1)\end{aligned}$$

From figure 4.2

$$\begin{aligned}\sum F_x &= 0 \\ -T + \mu_{s1} N_1 + \mu_{s2} N_2 &= 0 \\ -T + 0.5(98) + 0.6(294) &= 0 \dots \dots \dots (2) \\ T &= 225.4 \text{ N}\end{aligned}$$

Putting the value of T in equation 1

$$\begin{aligned}P - 225.4 - 0.5(98) &= 0 \\ P &= 274.4 \text{ N}\end{aligned}$$

Case b: When force P is applied on 20 kg block

Step 4: Free body diagram for case b

[1 mark]

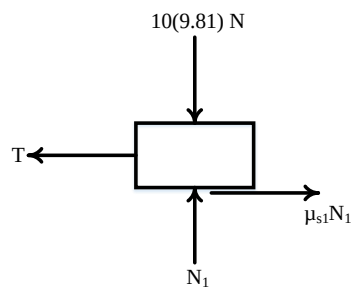


Figure 4.3

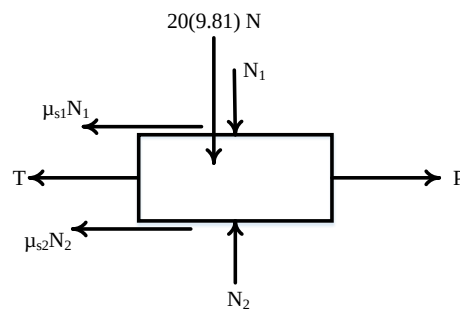


Figure 4.4

Step 5: Applied force P on 20 kg block and Tension T in the cable for case b [3 marks]

Force balance equation in x -direction, taking right direction force is positive and left direction force is negative. From figure 4.3

$$\begin{aligned}\sum F_x &= 0 \\ -T + \mu_{s1}N_1 &= 0 \\ -T + 0.5(98) &= 0 \\ T &= 49 \text{ N}\end{aligned}$$

From figure 4.4

$$\begin{aligned}\sum F_x &= 0 \\ -T - \mu_{s1}N_1 - \mu_{s2}N_2 + P &= 0 \\ -T - 0.5(98) - 0.6(294) + P &= 0 \quad \dots \dots \dots (3)\end{aligned}$$

Putting the value of T in equation 3

$$\begin{aligned}-49 - 0.5(98) - 0.6(294) + P &= 0 \\ P &= 274.4 \text{ N}\end{aligned}$$

Q5. Figure 5 shows a 5-kg uniform serving table being supported on each side by pairs of two identical links, AB and CD , and springs CE . The bowl has a mass of 1 kg. The springs each have a stiffness of $k = 200 \text{ N/m}$ and are unstretched when $\theta = 90^\circ$. Neglect the mass of the links.

- i) Draw the Free Body Diagram. [2 marks]
- ii) Write the spring force expression in terms of angle θ . [1 mark]
- iii) Write the virtual displacement expressions for the loads. [3 marks]
- iv) Write the virtual work equation. [1 mark]
- v) Determine the angle θ where the table is in equilibrium. [2.5 marks]
- vi) Determine the spring force when the table is in equilibrium. [0.5 marks]

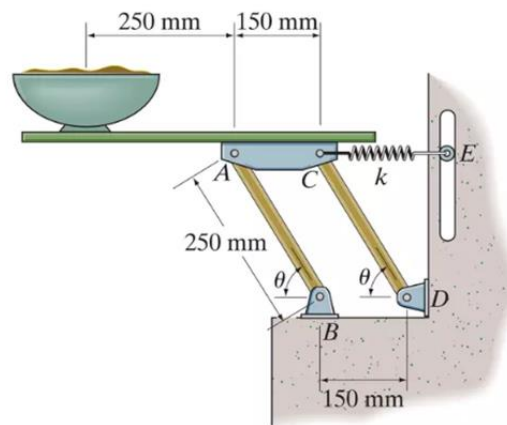


Figure 5

Solution of Q5:

Step 1: Drawing the Free-Body Diagram:

[2 marks]

When θ undergoes a positive virtual angular displacement of $\delta\theta$, the dash line configuration shown in **Figure 5.1** is formed. We observe that only the spring force $\mathbf{F}_{sp}$, the weight $\mathbf{W}_t$ of the table, and the weight $\mathbf{W}_b$ of the bowl do work when the virtual displacement takes place.

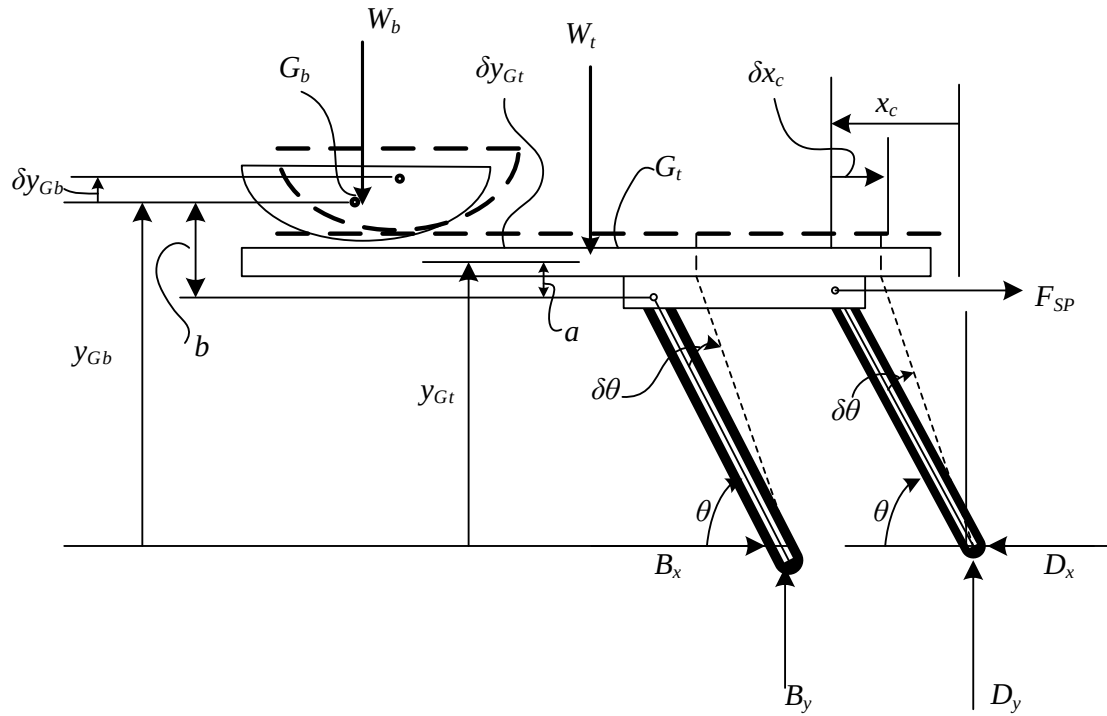


Figure 5.1

Step 2: Writing the spring force expression:

[1 mark]

The magnitude of $\mathbf{F}_{sp}$ can be computed using the spring force formula,

$$F_{sp} = kx = 200(0.25\cos\theta) = 50\cos\theta \text{ N}$$

Step 3: Writing the Virtual Displacement expressions for the loads:

[3 marks]

The position of points of application of $\mathbf{W}_b$, $\mathbf{W}_t$, and $\mathbf{F}_{sp}$ are specified by the position coordinates y_{Gb} , y_{Gt} , and x_c , respectively. Here, y_{Gb} and y_{Gt} are measured from the fixed point B while x_c is measured from the fixed point D .

$$\begin{aligned} y_{Gb} &= 0.25\sin\theta + b & \delta y_{Gb} &= 0.25\cos\theta\delta\theta \\ y_{Gt} &= 0.25\sin\theta + a & \delta y_{Gt} &= 0.25\cos\theta\delta\theta \\ x_c &= 0.25\cos\theta & \delta x_c &= -0.25\sin\theta\delta\theta \end{aligned} \quad \text{-----Eq. (1)}$$

Step 4: Writing the virtual work equation:

[1 mark]

Since $\mathbf{W}_b$, $\mathbf{W}_t$, and $\mathbf{F}_{sp}$ act towards the negative sense of their corresponding virtual displacement, their work is negative. Thus,

$$\delta U = 0; \quad -W_b\delta y_{Gb} + (-W_t\delta y_{Gt}) + (-F_{sp}\delta x_c) = 0 \quad \text{-----Eq. (2)}$$

Step 5: Determining the angle θ where the table is in equilibrium**[2.5 marks]**

Substituting $W_b = \left(\frac{1}{2}\right)(9.81) = 4.905 \text{ N}$, $W_t = \left(\frac{5}{2}\right)(9.81) = 24.525 \text{ N}$,

$F_{sp} = 50\cos\theta \text{ N}$, Eq. (1), into Eq. (2), we have

$$\begin{aligned} -4.905(0.25\cos\theta\delta\theta) - 24.525(0.25\cos\theta\delta\theta) - 50\cos\theta(-0.25\sin\theta\delta\theta) &= 0 \\ \text{Or, } \delta\theta(-7.3575\cos\theta + 12.5\sin\theta\cos\theta) &= 0 \end{aligned}$$

Since $\delta\theta \neq 0$, then

$$\begin{aligned} -7.3575\cos\theta + 12.5\sin\theta\cos\theta &= 0 \\ \cos\theta(-7.3575 + 12.5\sin\theta) &= 0 \end{aligned}$$

Solving the above equation,

$$\cos\theta = 0$$

$$\theta = 90^\circ,$$

it is corresponding to unstable equilibrium.

$$-7.3575 + 12.5\sin\theta = 0$$

$$\theta = 36.1^\circ$$

Step 6: Determining the spring force when the table is in equilibrium.**[0.5 marks]**

The spring force when the table is in equilibrium is given by,

$$F_{sp} = 50\cos\theta \text{ N} = 50 \times \cos 36.1^\circ = 40.4 \text{ N}$$